SPECS PDX DEV MEETUP

SwitchbackXR

The Switchback trail app on Spectacles, built in 3 days with gear from around the house

Spectacles ’24 · heart-rate strap · goggle HUD · e-paper camera

THE KIT

Five things from around the house

Spectacles '24

The lookout. Runs the lens and talks to everything else over Bluetooth.

Heart-rate strap

Scosche Rhythm 24 armband, standard heart-rate service. No SDK.

goggl1

A ski goggle with a 128×64 transparent OLED on an ESP32-C3 Beetle.

maplet

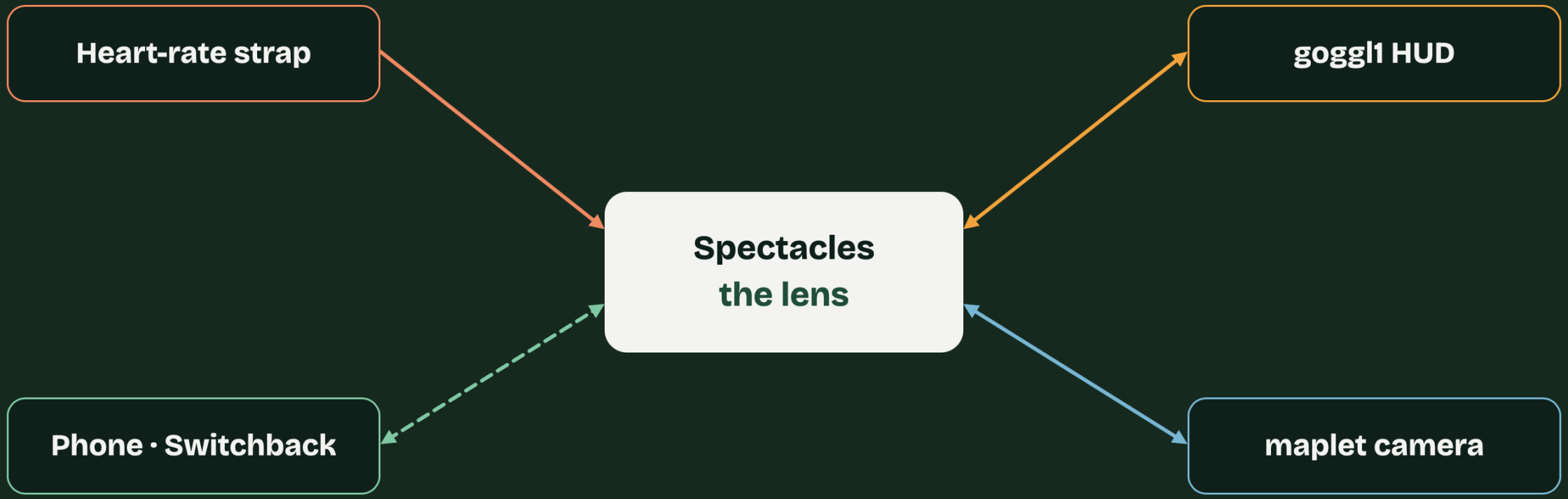
XIAO ESP32-S3 Sense: camera, microSD and a 2.13″ e-paper panel.

A phone

Switchback for Android: basecamp for GPS, maps and plans.

HOW IT HANGS TOGETHER

The glasses are the hub



Three Bluetooth links live at once. Phone link (dashed) is next: real GPS for the pin board.

THE ONE RULE

No bonding. No internet. Just Bluetooth.

~100 ms
round trip, lens to device

3
links at once, up within 4 s

0
pairing codes, clouds or SDKs

1 · PULSE

Your heart rate, on the glasses and the goggle

- The strap streams straight into the lens
- Beats per minute, zone and time on a floating panel
- The lens relays it to goggl1's Pulse screen
- It also sets the goggle's clock on connect

```
strap → lens

0x180D / 0x2A37
standard heart rate

lens → goggl1

sess-hr!
clock! hud-now
```

2 · REMOTE

Drive the goggle from the glasses

- goggl1's screens appear as true-pixel 128×64 cards
- Pinch a card and the goggle switches to it
- Switch its theme from the same panel

a pinch sends

2 view!

On the glasses, black is clear glass, so each card is just the goggle's lit pixels floating in the room.

3 · ROLL

The maplet's photos, pulled off its card

- The newest six frames, as a 1-bit contact sheet
- Amber paper, ink as clear glass
- Pinch a thumbnail to bring it up large

per 212×104 frame (2,808 B)

hex: 8–11 s
base64: 6–8 s

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ask for a faster link

6 12 ble-conn!
30 ms → 15 ms

4 · PIN BOARD

A topo map on the table, flags in the room

- A tilted contour plate with your track and pins
- Pin now, or auto-pin every 1, 5 or 15 minutes
- Flags stand in the room on each pin's real bearing
- Farther pins fly higher; labels step over each other

each flag reads

Summit

1.4 km · +312 m

Today a demo hiker walks the route.
Next: real GPS from the phone.

5 · SHOOT AND SEND

Take a photo with the glasses, print it on e-paper

Aim

A head-locked target at the e-paper's shape

Shoot

Only what's inside the target, mapped into the camera image

Dither

Grey, contrast-stretched, Floyd–Steinberg to 1 bit

Send

Drawn on the maplet and saved to its roll

The permission trick: Bluetooth plus camera is refused. Declare the Internet module too and it becomes a Transparent Permission lens: one prompt, then it works.

HOW IT WAS BUILT

The devices are Forth prompts. The lens just types.

Both boards run ESPIDFORTH, with the prompt over a Bluetooth serial link. A new feature is often one new word, tried live before it's flashed.

```
added this week
\ goggl1 0.6.1
1792700000 -25200 clock!

\ maplet 0.3.2
3 roll-b64
6 12 ble-conn!

\ the e-paper canvas
s" ff00…" epd-hex! epd-show
```

WHAT WE LEARNED

Six Spectacles gotchas, so you skip them

What happened	What fixed it
Lens Bluetooth can't bond; MTU stays 23	No encryption on our devices; write in 20-byte chunks
WebView plus Bluetooth: denied	Keep the WebView package out of a Bluetooth lens
Camera plus Bluetooth: denied	Declare the Internet module: one prompt, then allowed
Images flickered on the panels	Z-fighting with the frame's glass; sit 3–4 mm in front
Panels grabbed pinches from far away	Container frame border was 50 cm; set it to 4
Log says "InternalError: interrupted"	That's a hung script being killed, not a restart

THREE DAYS

From “can it even connect?” to photos on e-paper

Day 1

Bluetooth without bonding, proven. Heart rate on the glasses and the goggle.

Day 2

Remote, Roll and the pin board. New firmware for goggl1 and the maplet.

Day 3

Shoot with the glasses, send to e-paper. Polish from testing on our own glasses.

5
views in the lens

3
live Bluetooth links

2
firmware releases

~2,650
lines of lens TypeScript

WHAT'S NEXT

- Real GPS from the phone for the pin board
- Faster sends: a maplet word that doesn't echo
- Shots straight onto the goggle's screen

Thanks!

Questions? Try it on after the talk. · [name / contact]